



Quantum-Enhanced Multi-Modal Vector Search

A Hybrid Database Web App: Classical vs. Quantum Similarity

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We run the same text-to-image query through seven engines: brute force, FAISS flat, FAISS HNSW, swap test, Grover, Grover with quantum state preparation, and a HNSW + swap-test hybrid. We compare retrieval accuracy (MRR) with quantum resource cost (qubits, circuit depth, oracle calls), plus one small IBM hardware validation. The goal is honest measurement, not a quantum-advantage claim.

1. Problem & Research Questions

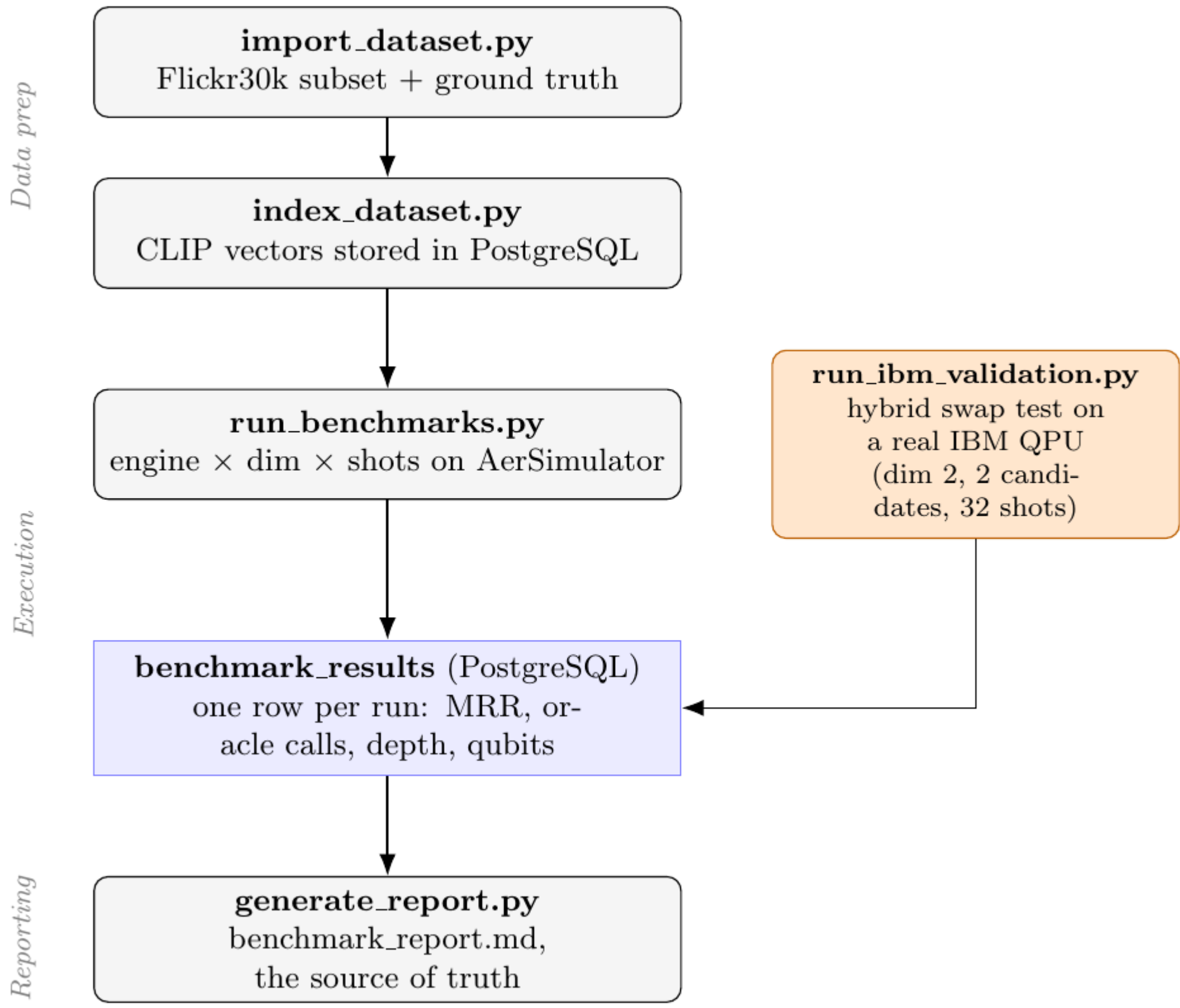
- Accuracy.** Does the swap-test reproduce classical cosine ranking, and at what shot budget?
- Scaling.** Does our Grover implementation match the $\lceil \pi\sqrt{N} \div 4 \rceil$ oracle-call curve on real circuits?
- Practicality.** Can a quantum engine beat a production HNSW baseline?
- Suitability.** Is vector retrieval a good problem class for quantum computing?

2. Why This Matters

Vector search supports recommendation, retrieval, and semantic search. Quantum algorithms promise asymptotic speedups, but few experiments measure them on real embeddings beside a fair classical baseline. This project builds that comparison now.

We are not claiming quantum is faster. Classical wins today. The contribution is the honest, reproducible measurement.

3. Methodology



Pipeline: Flickr30k images → CLIP ViT-B/32 embeddings → 64 / 128 / 256 dimensions → seven engines → PostgreSQL benchmark rows → FastAPI + React results.

- Dataset.** Flickr30k subset, N = 20 images, 20 text queries with hand-checked ground truth.
- Embeddings.** CLIP ViT-B/32, evaluated at 64 / 128 / 256 dimensions (plus dim 2 for hardware).
- Engines.** Eight in total: 3 classical, 3 quantum, 1 classical-quantum hybrid, 1 IBM QPU validation.
- Scale.** 920 benchmark runs across engine × dimension × query, one row each in PostgreSQL.
- Backends.** Qiskit AerSimulator for the sweep; one dim-2 run on a real IBM QPU at 32 shots.
- Reproducible.** Every number traces to benchmark_report.md, the generated source of truth.

4. Search Engines

Engine	Class	Complexity
brute_force_cosine	Classical	$O(N)$
faiss_flat_l2	Classical	$O(N)$
faiss_hnsw_l2	Classical	$O(\log N)$
qiskit_swap_test	Quantum	$O(N)$
qiskit_grover	Quantum	$O(\sqrt{N})$
qiskit_grover_quantum_prep	Quantum	$O(\sqrt{N})$
hybrid_hnsw_swap_test	Hybrid	$O(\log N + M)$
hybrid_hnsw_swap_test_ibm	IBM QPU	validation only

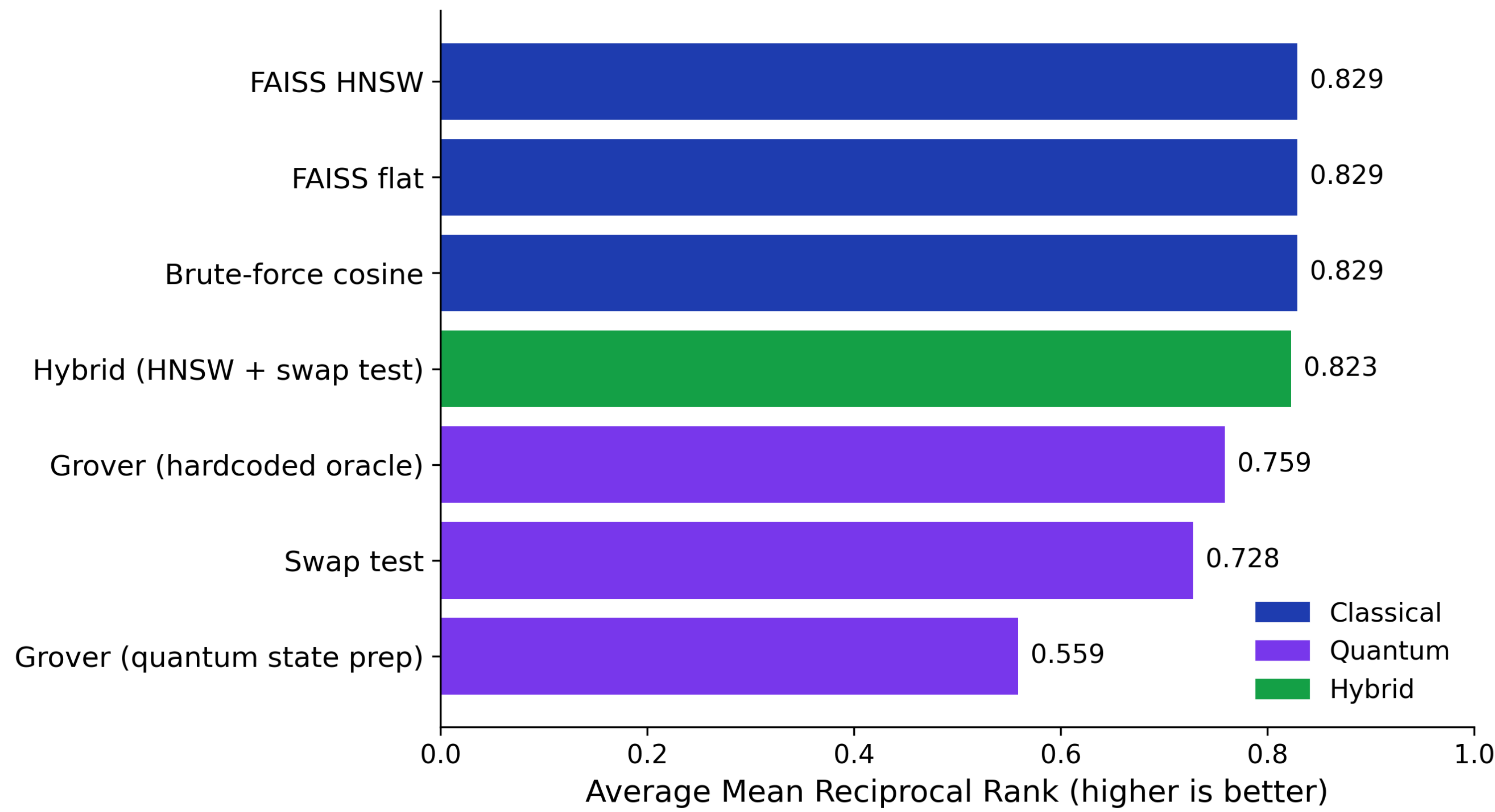
5. Evaluation Metrics

Mean Reciprocal Rank (MRR). Average of $1 / \text{rank}$ of first relevant result across queries. Higher is better; 1.0 means always first.

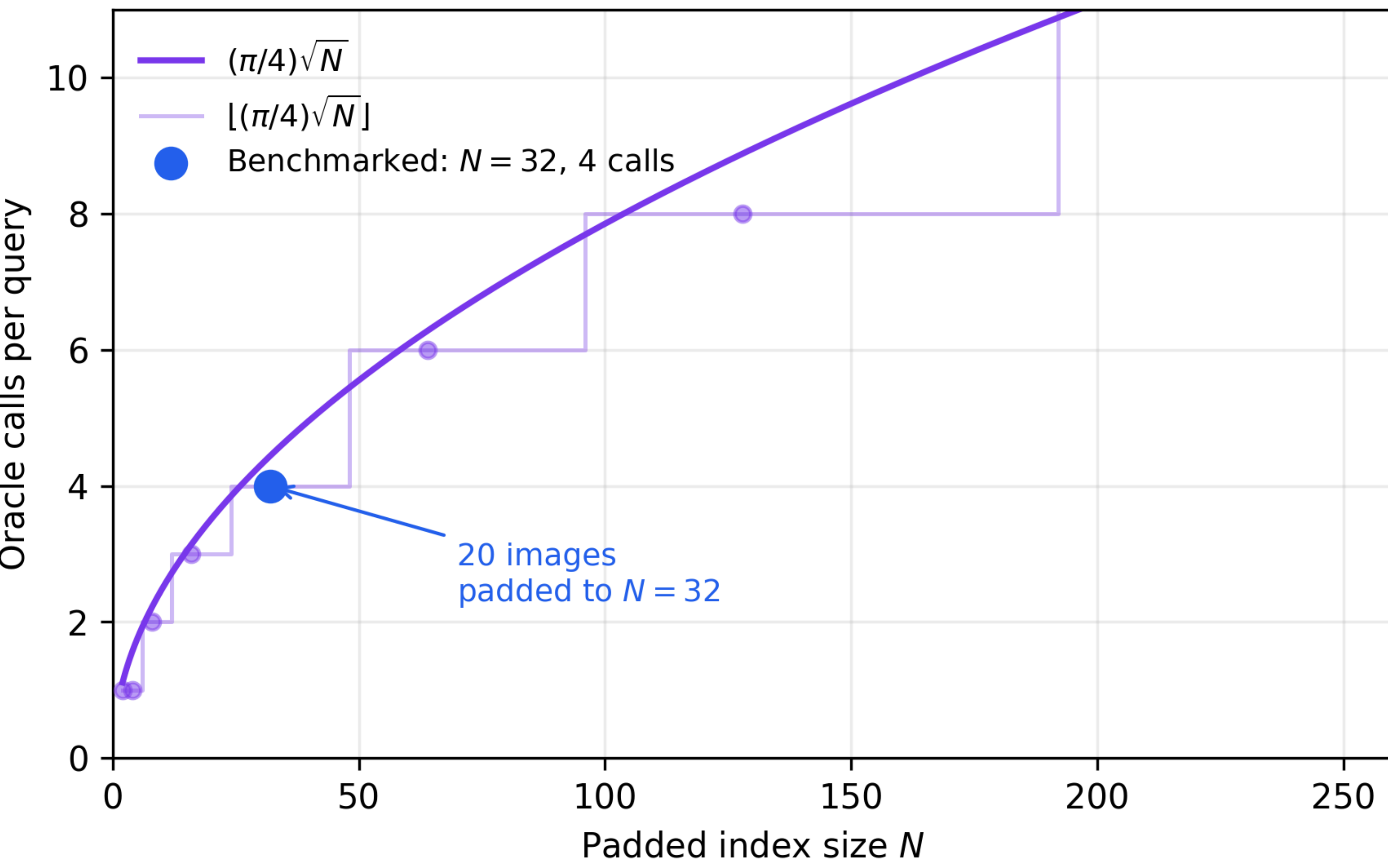
Oracle calls. Number of similarity comparisons per query. N for exact classical and swap test; $\lceil \pi\sqrt{N} \div 4 \rceil$ for Grover. Hardware-independent - the only valid cross-engine scaling metric.

No cross-engine wall-clock ranking: simulator time reflects classical CPU simulation overhead, not real quantum runtime.

6. Results



Classical engines tie at MRR 0.829; the hybrid is close at 0.823. The swap test reaches the classical ceiling at dimension 256. Accuracy is governed by embedding dimension, not by the quantum label.



Grover matches the theoretical $\lceil \pi\sqrt{N} \div 4 \rceil$ oracle count at the single padded index size tested (N = 32): one confirmed point, not the full curve.

7. The Honest Quantum Picture

Grover's $\sqrt{N}$ speedup is real for the *search step alone*. Loading N vectors without practical qRAM costs $O(N)$, cancelling the system-level speedup.

Even with ideal qRAM, classical HNSW achieves $O(\log N)$, which grows slower than $O(\sqrt{N})$. At $N = 10^6$, HNSW needs ~20 graph hops while Grover needs ~785 oracle calls. **HNSW wins on a laptop today.**

8. Conclusion & Future Work

- Benchmark harness** for classical vs. quantum vector search, controlled by one YAML file.
- Measured results** showing swap-test ranking recovery and Grover's $\sqrt{N}$ oracle count.
- Practical conclusion:** quantum vector search is not competitive today; the data-loading problem is central.

Next: larger dataset, full 512-dim CLIP, IBM hardware noise study, pgvector HNSW as a third classical baseline.

9. Key References

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